

Fault Diagnosis Algorithm for Biswapped Networks under the PMC Model *

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Abstract

Diagnosis is an essential subject for reliability of an interconnection network. This paper addresses the fault diagnosis of so called Biswapped networks[26], $Bsw(\Omega)$, which are related to OTIS networks[28] under the PMC model. Based on the cycle decomposition of $Bsw(\Omega)$, we design a diagnosis algorithm that can identify $t + 1$ faulty vertices of $Bsw(\Omega)$ if the basis network Ω satisfies all following conditions: 1) t -connected, 2) $|V(\Omega)| = 2i + 2$ for $i \geq t$, 3) Hamiltonian, and 4) triangle-free. By applying our algorithm, the faulty vertices of a Biswapped hypercube network can be identified in linear time.

Keywords: Interconnection Networks, Fault diagnosis algorithm, PMC model, Biswapped networks, OTIS networks.

1 Introduction

Multiprocessor systems are vulnerable to faults and fault-tolerant computing has become very important for systems with a large number of processors linked together by an interconnection network. In order to maintain the reliability of a system, a processor should be replaced by a fault-free processor whenever it is found faulty. The process of finding faulty processors is called fault diagnosis of the system. The degree of diagnosability of

the system denotes the maximal number of faulty processors that can be ensured to identify in the system.

System-level diagnosis is an important technique in improving the system reliability and availability. A graph-theoretical model of system-level fault diagnosis termed PMC model was proposed by Preparata, Metze and Chien [23]. In this model, every processor in a system performs tests on its neighbours via communication links between them. When a processor tests another one, the tester declares the tested processor to be fault-free or faulty depending on the test response and the result is always accurate if the tester is fault-free, but if the tester is faulty, the result is unreliable. A system is t -diagnosable if all faulty processors within the system can be identified when the number of faulty processors present does not exceed t . In [15], Hakimi and Amin gave a necessary and sufficient condition for verifying if a system is t -diagnosable under the PMC model. Dahbura and Masson [7] proposed general diagnosis algorithms with time complexity $O(N^{2.5})$ to identify all the faulty nodes in an N -node network.

Biswapped networks, proposed by Xiao et al.[26], are related to swapped or OTIS networks that are studied by a number of researchers [2, 5, 9, 10, 20, 21, 22, 28]. The Biswapped network, $Bsw(\Omega)$, is built of $2n$ copies of some n -node basis network, Ω , using a simple rule for connectivity that connected two nodes in different clusters by swapping the cluster labels and node labels. That can ensure its regularity, modularity, fault tolerance, and algorithmic efficiency. In particular, if Ω is a Cayley graph then so is the resulting $Bsw(\Omega)$. In this paper, we introduce a simple diagnosis algorithm to identify all the faulty vertices

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in the Biswapped networks with respect to utilizing the cycle decomposition. A cycle decomposition of $Bsw(\Omega)$ can be constructed by a Hamiltonian cycle of the basis network Ω . Suppose the time complexity to construct a Hamiltonian cycle in basis network is linear, then the time complexity of our algorithm is also linear. Hence, by applying our algorithm, Biswapped hypercube networks can be diagnosed in linear time.

In the next section, we introduce definitions and notations which are used in this paper. We then propose our diagnosis algorithm in Section 3. Finally, we give a conclusion in Section 4.

2 Preliminary

A multiprocessor system is conveniently represented with an undirected graph. The vertices (edges) of the graph represent the nodes (links) of the network. Throughout this paper, we focus on an undirected graph without loops and follow [3] for the graph definitions and notations. $G = (V, E)$ is a graph if V is a finite set and E is a subset of $\{(u, v) \mid (u, v) \text{ is an unordered pair of } V\}$. We say that V is the vertex set and E is the edge set. We also use $V(G)$ ($E(G)$) to represent the vertex (edge) set of a graph G . Two vertices u and v are adjacent if $(u, v) \in E$; the vertex u is called a neighbour of v , and vice versa. A path is a sequence of vertices in a graph of the form $\langle u_0, u_1, \dots, u_m \rangle$ such that u_i and u_{i+1} are adjacent for $i = 0, \dots, m-1$. In a simple path all the u_i are distinct. The *length* of a path P , denoted by $len(P)$, is the number of edges in P . A cycle is a simple path with $u_0 = u_m$. A graph is *triangle free*, if it has no cycle of length 3. A *cycle decomposition* is a partitioning of the vertices of a graph into subsets, such that the vertices in each subset lie on a cycle. A cycle is a *Hamiltonian cycle* if it traverses every vertex of G exactly once. A graph is *Hamiltonian* if it has a Hamiltonian cycle.

Under the PMC model, a system G is modelled by a digraph $T(V, E)$ in which an edge directed from vertex u to vertex v , denoted by $\langle u, v \rangle$, means that u tests v . In this situation, u is a tester and v is a tested vertex. A test of u to v is possible if these two nodes are connected in the system. A test result is a function $\sigma : E \rightarrow \{0, 1\}$ such that if u evaluates v as fault-free, $\langle u, v \rangle = 0$; otherwise, $\langle u, v \rangle = 1$. The result $\langle u, v \rangle$ is always accurate if u is fault-free, but if u is faulty, the result is unreliable. The collection of all test results is called a syndrome. The set of all faulty vertices in the sys-

tem is called a faulty set. For a given syndrome σ , a faulty set $F \subset V$ is called to be consistent with σ if, for a $\langle u, v \rangle \in E(T)$, such that $u \in V - F$, $\langle u, v \rangle = 1$ if and only if $v \in F$. A system is t -diagnosable if all faulty vertices can be identified provided that the number of faulty vertices does not exceed t . The maximum number of faulty vertices that the system can guarantee to detect is called the diagnosability of the system.

Let $\sigma(F)$ denote the set of all possible syndromes with which the faulty set F can be consistent. Then two distinct faulty sets $F_1, F_2 \subset V$ are said to be distinguishable if $\sigma(F_1) \cap \sigma(F_2) = \emptyset$; otherwise, F_1 and F_2 are said to be indistinguishable. Therefore, the following lemma is obtained.

Lemma 1 *A system G is t -diagnosable if and only if for any two distinct faulty set $F_1, F_2 \subset V(G)$ with $|F_1| \leq t$ and $|F_2| \leq t$, they are distinguishable.*

Lai et al. [18] give a sufficient and necessary condition to characterize t -diagnosable systems.

Lemma 2 *A system G is t -diagnosable if and only if for each $S \subset V(G)$ with $|S| = p$, $0 \leq p \leq t-1$, every connected component of $G - S$ has at least $2(t-p) + 1$ vertices.*

Biswapped networks are defined as follows. Given a simple graph Ω , the Biswapped network based on Ω , or called Biswapped Ω network, $Bsw(\Omega) = (V, E)$, is a graph with its vertex and edge sets specified as:

$$V = \{[i, p, g] \mid p, g \in V(\Omega) \text{ and } i \in \{0, 1\}\}$$

$$E = E_1 \cup E_2$$

$$E_1 = \{([i, p, g_1], [i, p, g_2]) \mid (g_1, g_2) \in E(\Omega) \text{ and } i \in \{0, 1\}\}$$

$$E_2 = \{([0, p, g], [1, g, p]) \mid p, g \in V(\Omega)\}$$

In the following, we use $[0, p]$, $[1, p]$ to denote the cluster number. The edge in E_1 is called *internal edge* and the edge in E_2 is called *outer edge*. The main properties of $Bsw(\Omega)$ have been investigated in [26]. We list the needed properties.

Property 1 *The degree of $Bsw(\Omega)$ is $deg(Bsw(\Omega)) = k + 1$ if the degree of Ω is $deg(\Omega) = k$.*

Property 2 *The connectivity of $Bsw(\Omega)$ is $k+1$ if the connectivity of Ω is k .*

Property 3 *Each vertex connects to only one outer edge.*

Property 4 $|V(Bsw(\Omega))| = 2n^2$ if $|V(\Omega)| = n$.

3 Fault diagnosis algorithm

In this section, we propose a cycle decomposition algorithm for $Bsw(\Omega)$ first. Then, we prepare some lemmas and a diagnosis algorithm on cycles for helping design the diagnosis algorithm.

Theorem 1 *Algorithm 1 can get a cycle decomposition with $|V(\Omega)|/2$ cycles of $Bsw(\Omega)$ if Ω is a Hamiltonian graph and $|V(\Omega)|$ is even.*

Proof. Let $n = |V(\Omega)|$ and $+$ be the operation of addition modulo n . To prove this theorem, we need to prove 1) $|V(Cd)| = |V(Bsw(\Omega))|$ and 2) any two edges in Cd are disjoint. In Algorithm 1, each edge on each cycle in Cd is labelled by $e_i^{[x,j]}$ where $e_i^{[x,j]} = (i, i+1)$ in cluster $[x, j]$. Without loss of generality, we assume that i is an even number. Then each cycle in Cd can be written as the following form: $e_i^{[0,0]} \rightarrow e_0^{[1,i+1]} \rightarrow e_{i+1}^{[0,1]} \rightarrow e_1^{[1,i+2]} \rightarrow e_{i+2}^{[0,2]} \rightarrow \dots \rightarrow e_{n-1}^{[1,i+n]} \rightarrow e_{i+n}^{[0,n]} = e_i^{[0,0]}$ for all even $0 \leq i \leq n-2$. By this form, the edges, except the outer edges, in a cycle are all in different clusters and the edges in a cluster are all in different cycles. Since there are $2n$ clusters, the length of each cycle is $4n$ and the number of cycles is $n/2$. Hence, $|V(Cd)| = 4n * n/2 = |V(Bsw(\Omega))| = 2n^2$. Then we claim that any two edges in Cd are disjoint. Let i, j be two different integers. Given two cycles start with $e_i^{[0,0]}$ and $e_j^{[0,0]}$, says C^i and C^j , in Cd . Clearly, the edges in different clusters are disjoint. Then we need to prove that any two edges in the same cluster but in different cycles are disjoint. Given a cluster number $[0, k]$, the edge in C^i is $e_{i+k}^{[0,k]}$ and the edge in C^j is $e_{j+k}^{[0,k]}$. Since $i \neq j$, $e_{i+k}^{[0,k]}$ and $e_{j+k}^{[0,k]}$ are disjoint in cluster $[0, k]$. Let $i+m = j+k$. Since $i \neq j$, $m \neq k$. Given a cluster number $[1, i+m] = [1, j+k]$. Then $e_{m-1}^{[1,i+m]}$ in C^i and $e_{k-1}^{[1,j+k]}$ in C^j are also disjoint in cluster $[1, i+m]$. Hence, Cd is a cycle decomposition of $Bsw(\Omega)$. $\square$

For example, $Cd = \{([0, 0, 0], [0, 0, 1], [1, 1, 0], [1, 1, 1], [0, 1, 1], \dots, [1, 0, 0], [0, 0, 0]) \text{ and } ([0, 0, 2], [0, 0, 3], [1, 3, 0], [1, 3, 1], [0, 1, 3], \dots, [1, 2, 0], [0, 0, 1])\}$ is a cycle decomposition of $Bsw(C_4)$. We illustrate the cycle decomposition in Fig. 1.

Corollary 1 *Algorithm 1 can run in $O(N)$ time complexity where N is the number of vertices of $Bsw(\Omega)$ if the Hamiltonian cycle of Ω can be found in $O(n)$ where $n = |V(\Omega)|$.*

Given a syndrome σ in a graph $G = (V, E)$, a test chain of order p , $W = \langle v_1, v_2, \dots, v_p \rangle$, is a directed path of distinct connected vertices along with a collection of test results. Let $V - V(W)$ denote the set of vertices $V - \{v_i \mid v_i \in W\}$. Consider the syndrome σ obtained from the test assignment in a test chain W . We call a maximal sequence of consecutive 0-arrows in the chain a σ -zero chain and we also call a maximal sequence of consecutive 1-arrows in the chain a σ -one chain. Therefore, any test chain are interleaved with σ -zero chains, denoted by $W_1^0, W_2^0, \dots, W_m^0$, and σ -one chains, denoted by $W_1^1, W_2^1, \dots, W_k^1$. It is observed that $|m-k| \leq 1$. The following two lemmas are helpful for recognizing the status of vertices in a graph.

Lemma 3 *Let $G = (V, E)$ be a graph with at most t faulty vertices, σ be a syndrome of G , and $W = \langle v_1, v_2, \dots, v_m \rangle$ be a σ -zero chain in G . If $V - V(W)$ contains k faulty vertices, then all v_i are fault-free for $i > t - k$.*

Proof. Let σ be a syndrome and $W = \langle v_1, v_2, \dots, v_m \rangle$ be a σ -zero chain in G . Suppose that there exists a faulty vertex $v_i \in V(W)$ where $i > t - k$. Since $\langle v_{i-1}, v_i \rangle = 0$, v_{i-1} is a faulty vertex, too. Thus, all v_j vertices must be faulty for $1 \leq j \leq i$. Hence the number of faulty vertices in W is more than $t - k$. Since $V - V(W)$ contains k faulty vertices, G has at least $t + 1$ faulty vertices which is contradiction. Therefore, all v_i in W are fault-free for $i > t - k$. $\square$

Lemma 4 *Let $G = (V, E)$ be a graph, σ be a syndrome of G , and $W = \langle v_1, v_2, \dots, v_m \rangle$ be a σ -one chain in G . Then, the set $V(W)$ contains at least $\lfloor \frac{m}{2} \rfloor$ faulty vertices.*

Theorem 2 *Algorithm 3 can identify the faulty vertex set F in $Bsw(\Omega)$ for $|F| \leq t + 1$ if the basis network Ω satisfies all following conditions: 1) t -connected, 2) $|V(\Omega)| = 2i + 2$ for $i \geq t$, 3) Hamiltonian, and 4) triangle-free.*

Proof. Based on the cycle decomposition algorithm, each cycle in Cd has only one edge in a cluster. Let $F(Cd)$ be the set of cycles in Cd such that each cycle has at least one σ -one chain. Since the number of faulty vertices $|F| \leq t + 1$, if $|F(Cd)| \geq t$ then each cycle in Cd has no more than 2 faulty vertices. Thus, we can use Algorithm 2 to identify the faulty vertices in each cycles. Then, we only need to consider that $|F(Cd)| \leq t - 1$. Clearly, we have $\frac{|V(\Omega)|}{2} - (t - 1)$ cycles that are fault-free. Then we claim that each unrecognised vertex $u \in U$ has

at least one neighbour, v , in the same cluster is fault-free. Suppose that v does not exist. Then each neighbour of u is in U or F . Since the number of neighbours of u is t in a cluster, at least t vertices are in U or F and these t vertices are in different cycles. Since each vertex in U or F has a related σ -one chain, there are t cycles containing σ -one chains. Hence, $|F(Cd)| \geq t$. We get a contradiction. Therefore, we always can find a fault-free vertex to identify the unrecognised vertex u . Then line 23 to 30 in Algorithm 3 can recognise all vertices in U . The algorithm is correct. $\square$

Theorem 3 *The biswapped network, $Bsw(\Omega)$, is $t+1$ diagnosable under the PMC model if the basis network Ω is t -connected.*

Since the time complexity to construct a Hamiltonian cycle in hypercubes is linear. Based on our algorithm, we have the following corollary.

Corollary 2 *The biswapped hypercube network can be diagnosed in linear time.*

4 Conclusion

Under the PMC model, we propose a diagnosis algorithm to identify all faulty vertices on the Biswapped Networks, $Bsw(\Omega)$, if the basis network Ω is t -connected, has more than $2t+1$ even number of vertices, is Hamiltonian, and is triangle-free. The algorithm uses the cycle decomposition to help identify the faulty vertices. The cycle decomposition of $Bsw(\Omega)$ can be construct by a Hamiltonian cycle of Ω . Hence, if a Hamiltonian cycle can be found in linear time then our algorithm also can run in linear time. $Bsw(\Omega)$ is related to swapped or OTIS networks which are used in optical networks. This algorithm may help design a self-fault diagnosis system in OTIS networks.

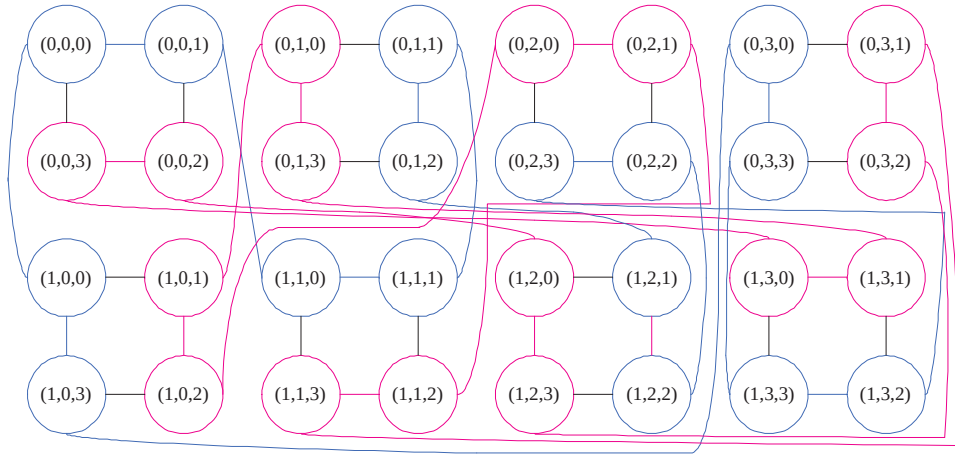
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Algorithm 1 A Cycle decomposition for $Bsw(\Omega)$ **Input:** A graph Ω which is Hamiltonian and $|V(\Omega)|$ is even.**Output:** A cycle decomposition with $|V(\Omega)|/2$ cycles of $Bsw(\Omega)$.

- 1: $n \leftarrow |V(\Omega)|$
- 2: Find a Hamiltonian cycle in Ω
- 3: Label each vertex in Ω from 0 to $n - 1$ by the vertex order on the Hamiltonian cycle, and label each edge by $e_i^{[x,j]}$ which represents the edge $(i, i + 1)$ in cluster $[x, j]$. $\{+ \text{ is the operation of addition modulo } n.\}$
- 4: **for** each even number i where $0 \leq i \leq n$ **do**
- 5: Add the cycle with the form: $e_i^{[0,0]} \rightarrow e_0^{[1,i+1]} \rightarrow e_{i+1}^{[0,1]} \rightarrow e_1^{[1,i+2]} \rightarrow e_{i+2}^{[0,2]} \rightarrow \dots \rightarrow e_{n-1}^{[1,i+n]} \rightarrow e_{i+n}^{[0,n]} = e_i^{[0,0]}$ into Cd , where $\rightarrow$ denotes an outer edge of $Bsw(\Omega)$.
- 6: **end for**
- 7: **return** Cd .

Figure 1: An illustration of a cycle decomposition of $Bsw(C_4)$.

Algorithm 2 A fault diagnosis algorithm for the cycle of n vertices, C_n , for $n \geq 5$.

Input: A cycle of n vertices, C_n , and $n \geq 5$.

Output: A set of all fault vertices, F , in C_n with $|F| \leq 2$.

```

1:  $F \leftarrow \phi$ .
2: Scan  $C_n$  from one direction to get test chains,  $Tc$ .
3: if  $Tc$  has only one  $\sigma$ -one chain, i.e.  $Tc = W_1^0, W_1^1$ . then
4:   Add first vertex in  $W_1^1$  into  $F$ .
5:   if  $|W_1^1| = 1$  then
6:     Use the second vertex,  $u$ , of  $W_1^0$  to test the first vertex,  $v$ , of  $W_1^0$ .
7:     if  $u$  says that  $v$  is faulty then
8:       Add  $v$  into  $F$ .
9:     end if
10:  else if  $|W_1^1| = 2$  then
11:    Use the first vertex,  $u$ , of  $W_1^0$  to test the last vertex,  $v$ , of  $W_1^1$ .
12:    if  $u$  says that  $v$  is faulty then
13:      Add  $v$  into  $F$ .
14:    end if
15:  else if  $|W_1^1| = 3$  then
16:    Use the first vertex,  $u$ , of  $W_1^0$  to test the last vertex,  $v$ , of  $W_1^1$ .
17:    if  $u$  says that  $v$  is faulty then
18:      Add  $v$  into  $F$ .
19:    else
20:      Add the second vertex of  $W_1^1$  into  $F$ .
21:    end if
22:  else if  $|W_1^1| = 4$  then
23:    Add the first vertex and the third vertex into  $F$ .
24:  else
25:    Output that there are more than two faulty vertices.
26:  end if
27: else if  $Tc$  has two  $\sigma$ -one chains, i.e.  $Tc = W_1^0, W_1^1, W_2^0, W_2^1$  and assume that  $|W_1^0| \geq |W_2^0|$ . then
28:   Add first vertex in  $W_1^1$  into  $F$ .
29:   Use the first vertex,  $u$ , of  $W_1^0$  to test the last vertex,  $v$ , of  $W_2^1$ .
30:   if  $u$  says that  $v$  is faulty then
31:     Add  $v$  into  $F$ .
32:   else
33:     Add the first vertex of  $W_2^1$  into  $F$ .
34:   end if
35: else
36:   Output that there are more than two faulty vertices.
37: end if
38: return  $F$ .

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Algorithm 3 A fault diagnosis algorithm for $Bsw(\Omega)$

Input: A syndrome σ of $Bsw(\Omega)$ where Ω satisfies all following conditions: 1) t -connected, 2) $|V(\Omega)| = 2i + 2$ for $i \geq t$, 3) Hamiltonian, and 4) triangle-free.

Output: A set of all faulty vertices, F , in $Bsw(\Omega)$ with $|F| \leq t + 1$.

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1:  $F \leftarrow \phi$ ,  $H \leftarrow \phi$ , and  $U \leftarrow \phi$ . { $F$  the faulty set,  $H$  the health set,  $U$  the unrecognised set.}
2:  $n \leftarrow |V(\Omega)|$ .
3: Call Algorithm 1, get the cycle decomposition,  $Cd$ , of  $Bsw(\Omega)$ .
4: Scan each cycle  $C^i$  in  $Cd$  from one direction to get test chains,  $Tc^i$ .
5: Divide each  $Tc^i$  into  $\sigma$ -zero chains  $W_1^{0,i}, W_2^{0,i}, \dots, W_s^{0,i}$  and  $\sigma$ -one chains  $W_1^{1,i}, W_2^{1,i}, \dots, W_s^{1,i}$ . {Note that,  $\sigma$ -zero chains and  $\sigma$ -one chains are interleaved in  $Tc^i$ .}
6: Let  $F(Cd)$  be the set of cycles in  $Cd$  such that each cycle has at least one  $\sigma$ -one chain.
7: if  $|F(Cd)| \geq t$  then
8:   Call Algorithm 2 for each cycle in  $F(Cd)$  to get the faulty set,  $F$ .
9:   return  $F$ .
10: end if
11: {In the following,  $|F(Cd)| \leq t - 1$ .}
12:  $q \leftarrow \sum_{i=1}^{(n/2)} \sum_{j=1}^s \lceil \frac{|V(W_j^{1,i})|}{2} \rceil$  {Ref. to Lemma 4.}
13: for each  $W_j^{0,i}$  do
14:   if  $|W_j^{0,i}| < (t + 1) - q$  then
15:     Add all vertices of  $W_j^{0,i}$  into  $U$ .
16:     Add all vertices of  $W_j^{1,i}$  into  $U$ .
17:   else
18:     Add first  $t - q$  vertices of  $W_j^{0,i}$  into  $U$ .
19:     Add the remaining vertices of  $W_j^{0,i}$  into  $H$ .
20:     Add first vertex of  $W_j^{1,i}$  into  $F$ .
21:     Add the remaining vertices of  $W_j^{1,i}$  into  $U$ .
22:   end if
23: end for
24: for each vertex  $u \in U$  do
25:   Find a neighbour of  $u$ , says  $v$ , where  $v \in H$ .
26:   Use  $v$  to recognise  $u$ .
27:   if  $v$  says that  $u$  is faulty. then
28:      $F \leftarrow F \cup \{u\}$ .
29:   else
30:      $H \leftarrow H \cup \{u\}$ .
31:   end if
32: end for
33: return  $F$ .

```